PROCEDURE

- A long cylindrical glass jar with markings is taken.
- Fill the glass jar with the given experimental liquid.
- Two points A and B are marked on the jar. The mark A is made well below the surface of the liquid so that when the ball reaches A it would have acquired terminal velocity V.
- The radius of the metal spherical ball is determined using screw gauge.
- The spherical ball is dropped gently into the liquid.
- Start the stop clock when the ball crosses the point A. Stop the clock when the ball reaches B and note down the time 't'.
- Note the distance between A and B and use it to calculate terminal velocity.
- Now repeat the experiment for different distances between A and B. Make sure that the point A is suitable for the ball to acquire terminal velocity.

OBSERVATIONS

To find Terminal Velocity:

S.No.	Distance covered by the spherical ball (d) (m)	Time taken (t) (s)	Terminal Velocity (V) d/t (m s^{-1})
		MEAN	

CALCULATION

Density of the spherical ball δ = _____ $kg\,m^{-3}$

Density of the given liquid π = _____ $kg\,m^{-3}$

Coefficient of viscosity of the liquid $\eta = \frac{2r^2 g(\delta - \sigma)}{9V}$ = _____ $\mathrm{N\,s\,m^{-2}}$

RESULT

The coefficient of viscosity of the given liquid by stoke's method is found to be

η = _____ $\mathrm{N\,s\,m^{-2}}$

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